



Number: _____

Name: _____

Teacher: _____

GOSFORD HIGH SCHOOL

HSC Mathematics 2012

Assessment Task 1

November 2011

TIME ALLOWED 65 MINUTES + 5 Minutes reading

- Show all working for questions worth more than 1 mark.
- An approved calculator may be used

Marks

	Functions	/20
	Calculus	/19
	Locus and the parabola	/17
	Total	/56

Question 1 Functions (20 Marks)**Marks**

a) Draw separate sketches of

(i) $y = |x + 2|$

2

(ii) $y = \frac{3}{x-2}$

2

(iii) $y = \sqrt{2-x}$

2

(iv) $y = 2^{-x}$

2

b) State the domain and range for each of these functions.

4

c) A circle of radius 5 units has its centre at the point $(-3, 4)$.

(i) Find its equation

2

(ii) Find the coordinates of the points where the curve cuts the y axis.

2

d) Graph the region in which $y < x^2 - 2$ and $x^2 + y^2 \leq 4$ both hold simultaneously.

4

Question 2 Calculus (19 Marks) Start a New Sheet of Papera) Evaluate $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$

2

b) Differentiate from first principles $f(x) = 8 - 7x - 2x^2$

3

c) Find the equation of the normal to the curve $y = (2x + 1)^4$ at the point where $x = -1$. Give your answer in the general form.

3

d) Differentiate:

(i) $\frac{x^2 - 2x + 1}{x}$

2

(ii) $\frac{1}{3x^2} + x\sqrt{x}$

2

Question 2 (continued)**Marks**

e) Differentiate and give answer in simplest form:

(i) $y = \frac{3x - 1}{(1 - 2x)^3}$

3(ii) Hence state the x coordinate(s) of any point(s) where the tangent to the curve is parallel to the x axis**1**

Differentiate and give answer in simplest form

(iii) $f(x) = x\sqrt{2x - 1}$

3**Question 3 (Locus and the Parabola) (17 Marks) Start a New Sheet of Paper**a) For the parabola $x^2 = 12y$ write down the coordinates of the focus and the equation of the directrix. **2**b) (i) Derive the locus of a point P which moves such that AP is perpendicular to PB if A is (3, 0) and B is (-1, 6). **3**(ii) Show that the locus is a circle and find its centre and radius. **2**c) For the parabola $y^2 = 8(3 - x)$ find: **4**

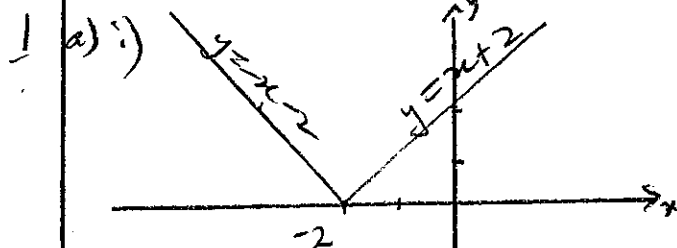
(i) The coordinates of the vertex

(ii) The coordinates of the focus

(iii) The equation of the directrix

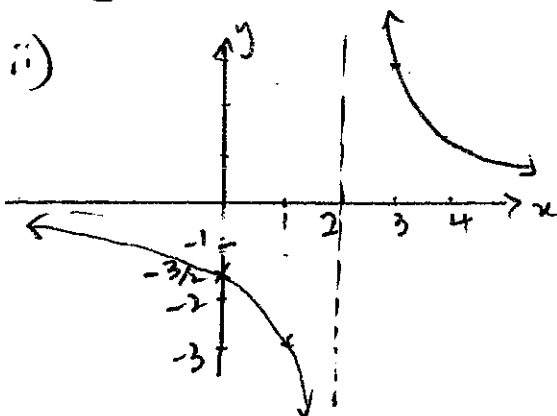
(iv) Sketch the parabola

d) (i) Derive the equation of the locus of a point P which is equidistant from the x axis and the line $4x + 3y = 12$. **3**(ii) Describe geometrically this locus and draw a sketch of it together with the original lines. Label each graph. **3**



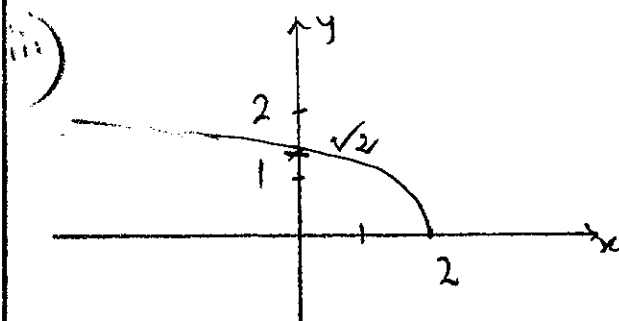
D: All real x

R: $y \geq 0$



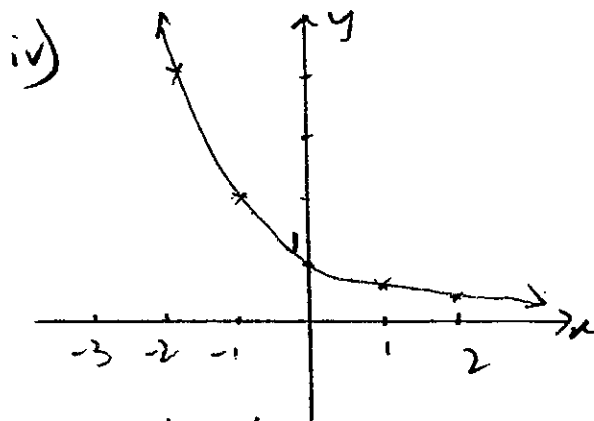
D: All real x except $x = 2$

R: All real y except $y = 0$



D: $x \leq 2$

R: $y \geq 0$



D: All real x

R: $y > 0$

c) i) $(x+3)^2 + (y-4)^2 = 25$

ii) y axis put $x = 0$

$$9 + (y-4)^2 = 25$$

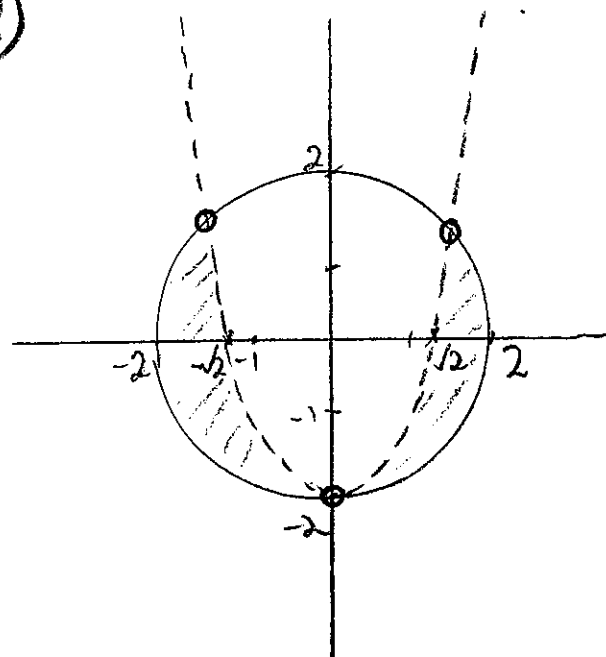
$$(y-4)^2 = 16$$

$$y-4 = \pm 4$$

$$y = 8 \text{ or } 0$$

$\therefore$ Points are $(0, 0) + (0, 8)$

d)



Test $(0, 0)$

$$0^2 + 0^2 \leq 4 \quad \checkmark$$

$\therefore$ Inside circle

$$0 < 0 - 2 \quad \text{False}$$

$\therefore$ Outside parabola

$$2) a) \lim_{x \rightarrow 4} \frac{(x-4)(x+4)}{x-4}$$

$$= \lim_{x \rightarrow 4} x+4 = 8$$

$$b) f(x) = 8 - 7x - 2x^2$$

$$f(x+h) = 8 - 7(x+h) - 2(x+h)^2$$

$$= 8 - 7x - 7h - 2x^2 - 4xh - 2h^2$$

$$f(x+h) - f(x) = -7h - 4xh - 2h^2$$

$$\frac{f(x+h) - f(x)}{h} = -7 - 4x - 2h$$

$$f'(x) = \lim_{h \rightarrow 0} -7 - 4x - 2h$$

$$= -7 - 4x$$

$$c) y = (2x+1)^4$$

$$x = -1, y = (-1)^4 = 1$$

$$y' = 4(2x+1)^3 \cdot 2$$

$$= 8(2x+1)^3$$

$$x = -1, m_1 = 8 \times (-1)^3$$

$$= -8$$

$\therefore$ For normal

$$m_2 = \frac{1}{8}$$

$$\therefore y - 1 = \frac{1}{8}(x + 1)$$

$$8y - 8 = x + 1$$

$$x - 8y + 9 = 0$$

$$d) i) y = x - 2 + x^{-1}$$

$$y' = 1 - x^{-2}$$

$$= 1 - \frac{1}{x^2} \text{ or } \frac{x^2 - 1}{x^2}$$

$$ii) y = \frac{1}{3}x^{-2} + x^{\frac{3}{2}}$$

$$y' = -\frac{2}{3}x^{-3} + \frac{3}{2}x^{\frac{1}{2}}$$

$$y' = \frac{-2}{3x^3} + \frac{3\sqrt{x}}{2}$$

$$e) i) y' = \frac{(-2x)^3 \cdot 3 - (3x-1) \cdot 3(1-2x)^2}{(1-2x)^6}$$

$$y' = \frac{3(1-2x)^2 \{1-2x + 2(3x-1)\}}{(1-2x)^6}$$

$$y' = \frac{3(4x-1)}{(1-2x)^4}$$

$$ii) y' = 0 \text{ when } x = \frac{1}{4}$$

$$iii) f(x) = x(2x-1)^{\frac{1}{2}}$$

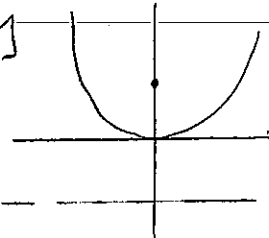
$$f'(x) = x \cdot \frac{1}{2}(2x-1)^{-\frac{1}{2}} \cdot 2 + (2x-1)^{\frac{1}{2}} \cdot 1$$

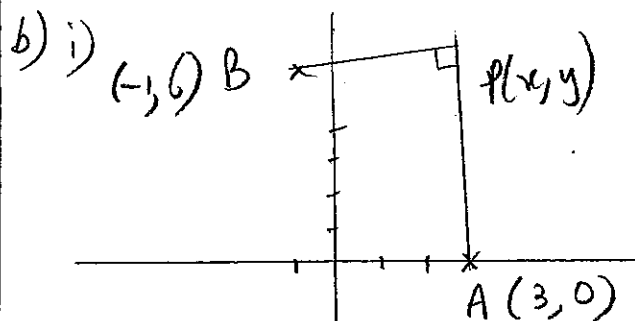
$$= \frac{x}{\sqrt{2x-1}} + \sqrt{2x-1}$$

$$= \frac{x + 2x - 1}{\sqrt{2x-1}}$$

$$= \frac{3x-1}{\sqrt{2x-1}}$$

3) a) $x^2 = 4 \times 3 \times y$
 S is (0, 3)
 D is $y = -3$





Use gradients or Pythagoras.

$$m_1 = \frac{y}{x-3}, \quad m_2 = \frac{y-6}{x+1}$$

$$\frac{y}{x-3} \cdot \frac{y-6}{x+1} = -1$$

$$y(y-6) = -(x-3)(x+1)$$

$$(x-3)(x+1) + y(y-6) = 0$$

ii) $x^2 - 2x - 3 + y^2 - 6y = 0$

$$x^2 - 2x + 1 + y^2 - 6y + 9 = 13$$

$$(x-1)^2 + (y-3)^2 = 13$$

This is in the form
 $(x-n)^2 + (y-y_1)^2 = r^2$
 which is a circle
 Centre (1, 3) radius $\sqrt{13}$

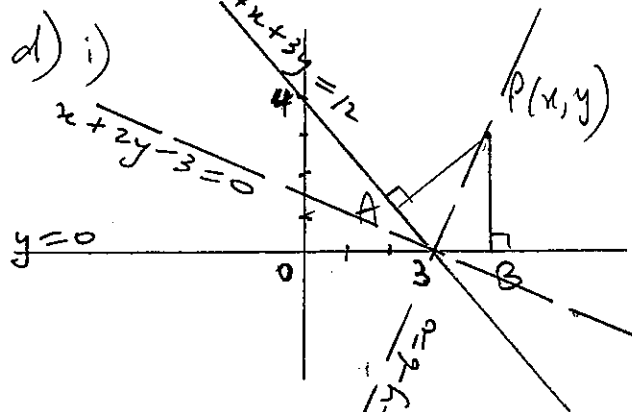
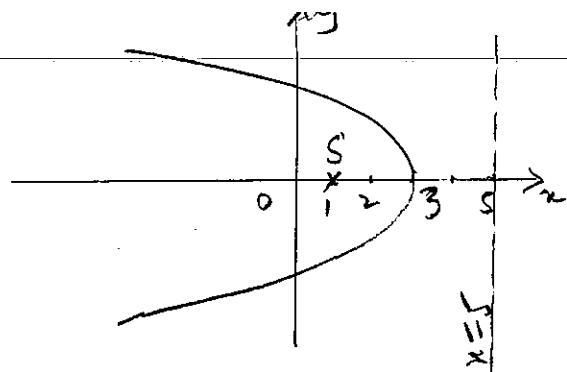
c) $y^2 = -8(x-3)$

$$(y-0)^2 = -4 \times 2(x-3)$$

i) Vertex is (3, 0)

ii) Focus is (1, 0)

iii) Directrix is $x = 5$



$$PA = PB$$

$$\left| \frac{4x+3y-12}{\sqrt{4^2+3^2}} \right| = |y|$$

$$\therefore 4x+3y-12 = 5y \quad \text{or}$$

$$4x+3y-12 = -5y$$

Equations are

$$4x-2y-12 = 0$$

$$\text{or } \underline{2x-y-6 = 0}$$

and

$$4x+8y-12 = 0$$

$$\text{or } \underline{x+2y-3 = 0}$$

ii) The locus is a pair of perpendicular straight lines, bisecting the angles between the x axis & $4x+3y=12$